

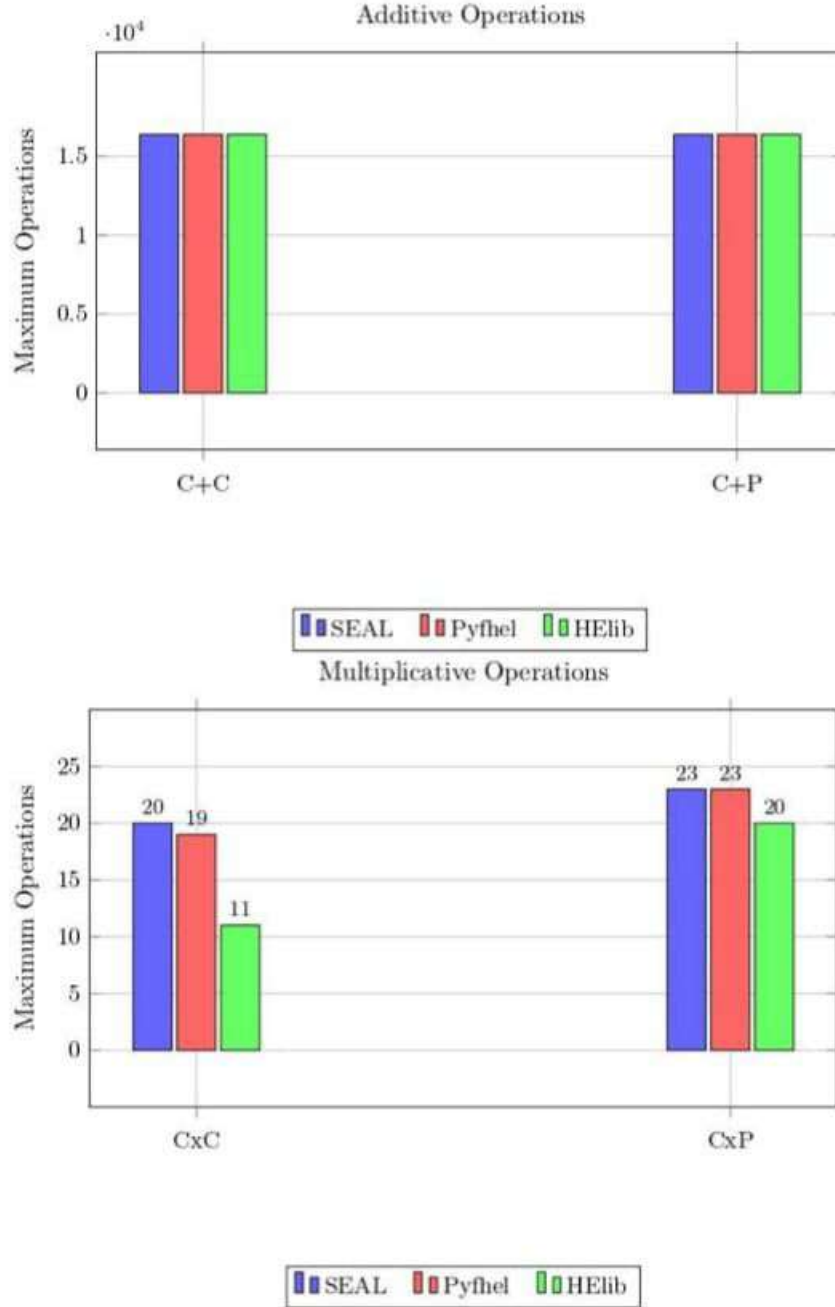


Figure 1: Comparison of maximum homomorphic operations at polynomial degree 32768. **Left:** Additive operations (C+C: Cipher+Cipher, C+P: Cipher+Plain) reach the safety cap of 16,384 for all libraries. **Right:** Multiplicative operations (CxP: Cipher×Plain, CxC: Cipher×Cipher) show performance differences: Pyfhel leads in CxC (19), SEAL and Pyfhel tie in CxP (23), and HELib achieves 11 (CxC) and 20 (CxP) operations.